Muhammad Ahmed

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EDUCATION

California State University, Long Beach **May 2027 (Expected)**
Master of Science in Computer Science – Emphasis: Artificial Intelligence, GPA: 4.0/4.0 *Long Beach, CA, USA*

National University of Computer and Emerging Sciences (FAST-NUCES) **Jul 2023**
Bachelor of Science in Computer Science (Dean's List: 5x Honoree, Top 5% of Cohort) *Lahore, Pakistan*
Relevant Coursework: Data Structures & Algorithms, Distributed Computing, Operating Systems, Databases, Computer Networks, Deep Learning

WORK EXPERIENCE

Chapman University **Feb 2026 – Present**
Graduate Research Assistant – Machine Learning *Orange, CA, USA*
- Engineered a Python (**NumPy**, **SciPy**, **pandas**) feature extraction pipeline for an NIH-funded study (n = 18 lab, n = 12 home via **MyWalkSentinel** iOS app), using adaptive median + 0.75 MAD thresholding to achieve ICC$(2,1)$ up to **0.805** vs. 3D motion capture and ICC$(3,14)$ up to **0.972**.
- Adapting pretrained inertial foundation models (**LIMU-X**, **HART**) in **PyTorch** and **TensorFlow**, fine-tuning transformer encoders on clinical stroke gait and 360° turning telemetry to predict fall risk across supervised laboratory and unsupervised home settings.

California State University, Long Beach – CECM Lab **Aug 2026 – Present**
Graduate Research Assistant – Computational Science & Numerical Optimization *Long Beach, CA, USA*
- Built an automated Python time-series ETL and Fast Fourier Transform (FFT) spectral pipeline across **202 sensor stations** and **30 climate scenarios**, automating 7-day linear interpolation and 99th-percentile spike filtering over multi-decade telemetry.
- Formulated a log-binned least-squares optimizer (**Balanced Log-RMSE**) across 7 nonlinear models to resolve **>98% high-frequency spectral skew** (12,608/12,865 FFT points), improving median $R^2_{\log}$ from **0.882** to **0.921** (**143/156** best fits, ΔAIC = 0.0).

Paismo **Dec 2024 – Jul 2025**
Software Engineer *Lahore, Pakistan*
- Architected a multi-tenant policy management microservice in **Node.js** adopted by **30+ enterprise organizations** to automate attendance and payroll workflows; mentored junior engineers through code reviews, system design guidance, and production debugging.
- Redesigned **PostgreSQL** database schemas and composite indexing strategies, cutting query latency by **40%** at **100K+ daily requests**.

xiQ, Inc. **Jul 2023 – Dec 2024**
Software Engineer *Lahore, Pakistan*
- Built enterprise B2B email campaign management services and REST APIs in **Django**, collaborating with stakeholders to deliver customer targeting and analytics workflows serving **8,000+ daily active users (DAU)**.
- Scaled asynchronous email workflow execution to **300K+ runs/month** by profiling production bottlenecks and implementing request batching, database connection pooling, and **Redis** caching.

Tintash **Jul 2022 – Sep 2022**
Software Engineering Intern *Lahore, Pakistan*
- Shipped production REST endpoints and relational data models with request validation, pagination, and standardized error contracts.

PROJECTS

Temporal Python Dependency Conflict Resolver | *Python, PyTorch, Graph Algorithms, Shortest-Path Search* **Sep 2026 – Oct 2026**
- Modeled **1,047,966 PyPI packages** as a directed temporal dependency graph; combined a neural compatibility classifier with a version-jump penalty and shortest-path graph search to resolve transitive dependency conflicts and generate safe upgrade sequences.

Sobel-Laplacian-SIMD | *C++, ARM NEON, SIMD Intrinsics, Cache Locality Optimization* **Nov 2025 – Dec 2025**
- Optimized Sobel and Laplacian edge-detection kernels in **C++** using **ARM NEON SIMD** intrinsics and cache-efficient memory locality improvements, achieving a **14x execution speedup** over scalar implementations.

RoboInk: Automated 3D Card Printing System | *Python, Django, AWS (S3, SQS, RDS)* **Jun 2024 – Aug 2024**
- Architected an event-driven print orchestration backend in **Django** on **AWS (S3, SQS, RDS)** to convert SVG inputs into validated G-code and schedule asynchronous execution, increasing print throughput to **20 jobs/hour**.

PUBLICATIONS & ACHIEVEMENTS

- **1st Author – IEEE COMPSAC (2026):** *"Toward Reliable LLM Code Generation: An Adaptive Routing Framework for Ambiguous Requirements"* (pp. 1953–1958); **Co-Author – IEEE COMPSAC (2026):** *"Agentic RAG for Sustainable ETL Architecture (AURA)"* (pp. 3286–3291; AST + 4-agent verification across 12 ETL repos / 413 functions, reaching **0.85 F1** in design-smell detection).
- **1st Author – World Physiotherapy Congress (2027):** *"From Laboratory to Home: Agreement and Repeatability of IMU-Derived Walking, Turning, and Sit-to-Stand Mobility Markers After Stroke"* (NIH Grant 1R15HD110941-01).
- **Honors & Awards: CSULB CECS Departmental Excellence Scholarship** (sole recipient in the department); **1st Place Winner, Paismo Hackathon 2025** (built an end-to-end embedding-based semantic resume-to-job matching engine).

TECHNICAL SKILLS

Languages & Performance: Python, C++, Java, TypeScript/JavaScript, SQL, Bash | ARM NEON SIMD, Cache Locality, Graph & Numerical Algorithms
Distributed Systems & Backend: Microservices, Event-Driven Architecture, High-Throughput ETL, REST APIs, Node.js, Django, Flask, Async Queues
Data Stores & Cloud Infra: PostgreSQL (Query/Index Optimization), Redis, Kafka, MongoDB, Elasticsearch, AWS (S3, SQS, RDS, EC2), GCP, Docker, CI/CD
AI/ML & Foundation Models: PyTorch, TensorFlow, Transformer Fine-Tuning, Multi-Agent RAG, LangGraph, Vector Search, AST Analysis, SciPy, NumPy